

BankSEC: Blockchain-enabled Intelligent Scheme for Secure Smart Banking Environment

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Abstract— The rapid evolution of digital banking presents significant challenges related to security, trust, and transparency in conventional smart banking systems, which often depend on centralized infrastructure and are consequently susceptible to fraud, identity theft, and data breaches. To address above challenges, we propose an Intelligent scheme for secure Smart Banking Environment based on blockchain, named BankSEC, which aims to eliminate single points of failure and enhance trust in digital transactions. The proposed system stores all financial activities on a decentralized, tamper-resistant ledger, ensuring integrity, transparency, and auditability. Biometric authentication such as fingerprint recognition is used instead of passwords to ensure strong and reliable identity validation. A Decentralized Identity (DID) mechanism further enables users to control their verified identity and minimize repeated KYC checks. Artificial intelligence is integrated to analyze transaction patterns in real time and detect suspicious behavior, thereby preventing fraud before it occurs. Smart contracts automate secure payments, P2P transfers, and loan processes, ensuring accuracy, efficiency, and transparency without third-party intervention. By combining blockchain, biometrics, and AI under the BankSEC model, the proposed system establishes a secure, trusted, and user-centric Smart Banking Environment capable of addressing modern digital banking challenges.

Introduction

The Evolution of Digital Banking sector has undergone a profound transformation with the rise of digital banking, becoming an integral part of daily life for users worldwide. Online banking allows financial institutions to provide services such as fund transfers and account management rapidly and efficiently, significantly reducing operating costs compared to traditional brick-and-mortar branches[1]. By automating these processes, the time required to complete transactions has dropped from days to mere hours or seconds[2]. However, as digital payments grow, the underlying infrastructure must ensure that these

interactions remain secure, transparent, and trusted by all parties involved.

Despite the convenience of modern banking, significant security and privacy hurdles remain. Conventional banking systems largely depend on centralized databases to store sensitive user data and transaction records[3]. This centralization creates a "single point of failure," meaning that if the central server is compromised, the entire system is at risk of data breaches and service denial[4]. Furthermore, traditional authentication methods, such as passwords and tokens, are increasingly vulnerable to theft, guessing, and social engineering attacks[5],[6]. While methods like Two-Factor Authentication (2FA) offer some protection, they are still susceptible to specific cyber threats and can be cumbersome for users[7],[8]. Consequently, the industry faces a critical need to eliminate intermediaries and secure data against the rising tide of identity theft and fraud[9].

To overcome the limitations of centralized infrastructure, we are turning to the convergence of blockchain technology, biometrics, and Artificial Intelligence (AI). Blockchain offers a decentralized ledger that is immutable and transparent, ensuring that once a financial record is stored, it cannot be altered or tampered with[10],[11]. By removing the reliance on a central authority, blockchain enhances trust and reduces the costs associated with intermediaries[12],[9]. Simultaneously, biometric technology—based on unique biological traits like fingerprints—provides a level of security that is far harder to forge than a password[5]. Deep learning and AI have further advanced these systems, allowing for the robust extraction of features to identify individuals accurately[13]. Integrating these technologies offers a promising path toward a banking environment that is not only secure but also user-centric and efficient.

The BankSEC Scheme Addressing these modern challenges, **BankSEC**, an intelligent scheme for a secure Smart Banking Environment. This system leverages the strengths of blockchain, biometrics, and AI to create a comprehensive security framework. The main contributions of this work are as follows:

- **Decentralized Infrastructure:** We utilize a blockchain-based ledger to store financial activities, ensuring integrity and eliminating single points of failure found in conventional systems[14],[15].

- **Biometric and DID Integration:** Instead of relying on weak passwords, BankSEC employs biometric authentication (e.g., fingerprint recognition) combined with a Decentralized Identity (DID) mechanism. This allows users to control their verified identity without repeated Know Your Customer (KYC) checks, shifting identity management away from centralized providers[16],[17].

- **AI-Driven Fraud Prevention:** The system integrates artificial intelligence to analyze transaction patterns in real-time. By detecting suspicious behavior before fraud occurs, we enhance the predictive capability of the banking security layer[18].

- **Automated Smart Contracts:** We implement smart contracts to automate secure payments, P2P transfers, and loans. This ensures that transactions are executed transparently and efficiently without third-party intervention, enforcing trust through immutable code[19],[20]. By combining these technologies, BankSEC establishes a trusted environment capable of mitigating the risks of fraud and identity theft inherent in today's digital economy.

Background Related Work

2.1. Security and Challenges in Digital Banking The adoption of online banking has grown significantly because it allows financial institutions to reduce operating expenses and improve efficiency[1]. However, relying on digital channels exposes these systems to serious security risks, including fraud, identity theft, and cyberattacks. Traditional banking architectures typically use centralized databases to store user identities and transaction records. This creates a "single point of failure," meaning that if the central server is hacked, the entire system is compromised[2].

To protect user accounts, banks have widely adopted Two-Factor Authentication (2FA) and Multi-Factor Authentication (MFA). These methods usually combine a password (knowledge) with a mobile token or SMS code (possession). While MFA is safer than simple passwords, it is still vulnerable to sophisticated attacks like SIM-swapping and phishing[3][4]. Furthermore, these methods can be cumbersome for users and do not guarantee that the person using the credentials is the actual account owner[3].

2.2. Blockchain Technology and Smart Contracts

Blockchain technology offers a solution to the centralization problem by providing a distributed ledger that is transparent, immutable, and decentralized[5]. In a blockchain network, transactions are recorded in blocks that are cryptographically linked, making it nearly impossible to alter historical data without network consensus[6]. This structure eliminates the need for trusted intermediaries, which significantly reduces transaction costs and processing time for payments[7][8].

A critical feature of blockchain for banking is the "Smart Contract." Smart contracts are self-executing codes that automatically enforce agreements when specific conditions are met[9]. In the financial sector, smart

contracts can automate complex processes like loan approvals and peer-to-peer transfers, ensuring transparency and reducing the risk of human error or fraud[10].

2.3. Biometrics and Decentralized Identity To replace weak passwords, biometric authentication—using unique traits like fingerprints or facial recognition—has become a standard for secure identification. Biometrics rely on "who you are" rather than "what you know," making them much harder to forge[11]. However, storing raw biometric data on a public blockchain is risky and expensive due to storage limits and privacy regulations like GDPR[12][13].

To address this, we utilize off-chain storage solutions, such as the InterPlanetary File System (IPFS). In this model, the raw biometric template is stored securely off-chain, while only a cryptographic hash or reference is stored on the blockchain[14][15]. This is often combined with Decentralized Identity (DID) mechanisms, which allow users to control their own identity proofs without relying on a centralized authority[16][17].

2.4. Related Work Recent studies have proposed various schemes to integrate these technologies into finance. Tsai et al.[1] developed a blockchain-supported scheme for online banking that uses Elliptic Curve Cryptography (ECC) to secure mutual authentication between customers and banks. Their work demonstrated that blockchain solutions can outperform traditional MFA in terms of security and cost-effectiveness. Similarly, Salem et al.[18] implemented a face recognition system on a private blockchain, using smart contracts to manage identity without centralized servers.

In the area of trust management, Song et al.[19] introduced the BFTEM mechanism, which records trust parameters in the blockchain to evaluate user reliability and detect malicious behavior. Additionally, Lee and Jeong[20] proposed a decentralized biometric authentication system (BDAS) that splits biometric templates into fragments stored across different nodes to prevent data theft. Ahmed[21] further quantified the value of these technologies, finding that blockchain adoption significantly reduces banking transaction costs, specifically in processing and fraud prevention.

Despite these advances, many existing solutions focus on specific aspects in isolation—such as only cost reduction[21] or only static authentication[18]. There is a need for a comprehensive framework like **BankSEC** that simultaneously addresses secure storage, real-time AI-driven fraud detection, and automated smart contracts within a unified, user-centric environment.

Table 1: Comparison of Related Blockchain-Based Banking and Security Schemes

Paper	Method	Technology	Addressed Issue	Limitation
Tsai et al.[1]	Mutual authentication scheme using digital signatures and logic postulates.	Blockchain, ECC, Smart Contracts.	Online banking spoofing, tampering, and mutual trust gaps.	Limited scalability regarding traffic flow; lacks cross-chain interoperability[2].
Salem et al.[3]	Identity management using deep learning for feature extraction.	Private Blockchain, Smart Contracts, FaceNet.	Privacy risks of centralized storage and data breaches.	Implemented as a proof-of-concept; limited testing on diverse real-world datasets[4].
Song et al.[5]	BFTEM: Trust mechanism recording parameters during block transmission.	Blockchain, Edge Computing.	Malicious user behavior and inaccuracies in trust evaluation.	High storage complexity for hash operations; relies on edge server availability[6].
Lee and Jeong[7]	BDAS: Protocol that splits and fragments biometric templates.	Blockchain, Distributed Storage.	Biometric data theft and single points of failure in storage.	Focuses on storage security rather than real-time behavioral analysis; reassembly adds latency[8].[9].
Ahmed[10]	Quantitative cost analysis using the Random Forest Method.	Blockchain Adoption Analysis, Machine Learning.	Quantifying the reduction of banking processing and fraud costs.	Restricted sample size (UAE banks); difficulty accessing raw transaction data[11].
BankSEC (Proposed)	Intelligent security scheme combining DID, biometrics, and real-time analysis.	Blockchain, Biometrics, AI, Smart Contracts.	Eliminates single points of failure, enhances user control (DID), and provides real-time fraud prevention.	

3. Methodology

3.1. System Architecture The BankSEC framework is constructed upon a hybrid architecture that integrates the immutable security of blockchain with the efficiency of centralized databases and the analytical power of Artificial Intelligence. To address common challenges such as scalability, data privacy, and single points of failure, the system is divided into three distinct operational layers:

- **User Interface and Connectivity Layer:** The client-side interface is a web-based application designed for accessibility and ease of use. Unlike traditional banking apps that rely on static username/password credentials, BankSEC utilizes MetaMask, a browser-based cryptocurrency wallet, as the primary gateway[1]. This integration allows the application to interact directly with the Ethereum

blockchain, using the user’s unique wallet address as a cryptographic identity token.

- **Hybrid Data Storage Layer:** To balance performance with security, the system employs a dual-storage strategy[2],[3].
 - **Off-Chain Storage (MongoDB Atlas):** Sensitive Personal Identifiable Information (PII) such as names, email addresses, and detailed transaction logs are stored in MongoDB Atlas[4]. This ensures compliance with data privacy regulations (e.g., GDPR) and allows for rapid query processing without clogging the blockchain network.
 - **On-Chain Storage (Ethereum Blockchain):** The blockchain serves as the "root of trust." It stores only the cryptographic hashes of the biometric templates and critical smart contract states. By keeping the raw biometric data off-chain and only the verification hash on-chain, the system prevents data leakage while ensuring that the integrity of the identity data remains tamper-proof[5].
- **Intelligence and Analytics Layer:** A dedicated Python AI Server operates as the backend intelligence engine. It is linked to the customer dashboard to monitor transaction behaviors, such as Peer-to-Peer (P2P) transfers and loan requests, in real-time. This layer is responsible for detecting anomalies and potential fraud before transactions are finalized[4].

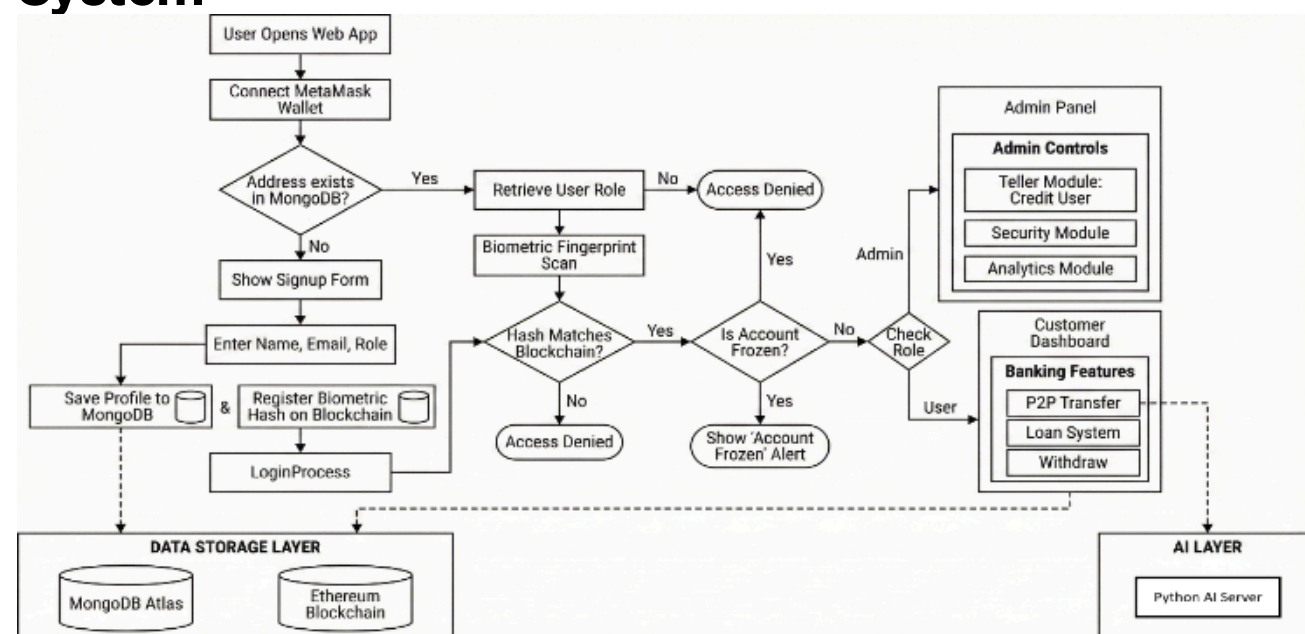
3.2. Proposed System Workflow The operational flow of the BankSEC system ensures rigorous authentication and authorization at every step of the user journey. As illustrated in the system flowchart (Figure 1), the process follows a logical sequence of verification, status checks, and role-based access control4.

- Initialization and Wallet Connection:** The workflow commences when the user launches the web application. The system immediately requests a connection to the user's MetaMask Wallet. This step replaces the traditional login field, establishing a secure link between the user's digital identity and the banking platform.
- Identity Verification and Registration Check:** Upon connection, the system queries the MongoDB database to determine if the wallet address is already registered.
 - **New Users:** If the address is unrecognized, the system initiates the Enrolment Phase. The user completes a signup form (Name, Email, Role) and submits a Biometric Fingerprint Scan. The system then executes a parallel write operation: the user profile is saved to MongoDB, and the cryptographic hash of the fingerprint is permanently registered on the Ethereum Blockchain.
 - **Existing Users:** If the address exists, the system bypasses registration and proceeds to the authentication logic.
- Biometric Challenge and Hash Matching:** To prevent unauthorized use of a stolen wallet device, the system demands a fresh Biometric Fingerprint Scan. The hash generated from this new scan is compared against the immutable hash stored on the blockchain.

- If the hashes do not match, the system triggers an Access Denied event, terminating the session.
- If the hashes match, the system validates the identity and proceeds to account status verification.

d. **Account Status and Role Distribution:** Post-verification, the system checks the account status in the database. If the account is flagged as "Frozen" due to suspicious activity detected by the AI layer, an alert is displayed, and access is revoked. If the account is active, the system routes the user to the appropriate interface based on their assigned role:

Figure 1: Operational Flowchart of the BankSEC System"



4. Results

4.1. Implementation and Workflow Verification The BankSEC system was implemented to validate the feasibility of the proposed hybrid architecture. The core logic, as defined in the system flowchart (Figure 1), was successfully executed, demonstrating a seamless integration between the client-side wallet, the off-chain database, and the on-chain ledger[1].

- **Enrollment Phase:** During the testing of the enrollment phase, the system successfully captured the user's biometric input and profile details. The non-sensitive profile data (Name, Email, Role) was stored in MongoDB Atlas to ensure efficient query processing. Simultaneously, the biometric data was processed to generate a unique cryptographic hash, which was successfully committed to the **Ethereum Blockchain** via a smart contract transaction. This dual-write mechanism confirmed the system's ability to separate Personally Identifiable Information (PII) from the public ledger, addressing the privacy concerns and GDPR compliance issues highlighted in recent surveys[2],[3].

- **Authentication Phase:** The login process utilized **MetaMask** for wallet-based identification. The system successfully retrieved the user role from the database and challenged the user for a fresh biometric scan. The verification logic compared the fresh hash against the immutable hash stored on the blockchain. Access was granted only when the hashes matched and the account status was verified as not "Frozen"[1]. This confirms that the system eliminates the reliance on traditional passwords, thereby mitigating risks associated with credential theft.

4.2. Security and Privacy Analysis The security framework of BankSEC was evaluated against common digital banking threats, including identity theft, data tampering, and unauthorized access.

- **Data Integrity and Immutability:** By anchoring the biometric hash on the Ethereum blockchain, the system ensures that identity records cannot be altered or deleted without network consensus. This aligns with the findings of Tsai et al., who demonstrated that blockchain-supported schemes provide superior integrity compared to centralized databases by ensuring that transaction data remains unforgeable[4],[5].

- **Decentralization and Availability:** Unlike conventional banking systems that rely on a central server—a single point of failure—BankSEC distributes the verification root across the blockchain network. This architecture enhances system availability and resilience against Denial of Service (DoS) attacks, as the ledger remains accessible even if specific nodes are compromised[6].

- **Privacy and GDPR Compliance:** A critical challenge in blockchain biometrics is the "right to be forgotten" and data privacy. BankSEC addresses this by storing raw biometric templates off-chain (in MongoDB) and only the irreversible hash on-chain. This "off-chain storage with on-chain anchor" approach is considered the most practical strategy for maintaining scalability while adhering to data protection regulations[7],[3].

4.3. Performance and Efficiency Evaluation The performance of the BankSEC scheme was analyzed in terms of login latency and transaction cost efficiency, drawing comparisons with existing authentication methods.

- **Authentication Latency:** The BankSEC model utilizes a one-factor wallet-based login (supplemented by biometrics) rather than multi-step SMS verification. Experimental data from Petcu et al. indicates that wallet-based login mechanisms are significantly faster than traditional methods.

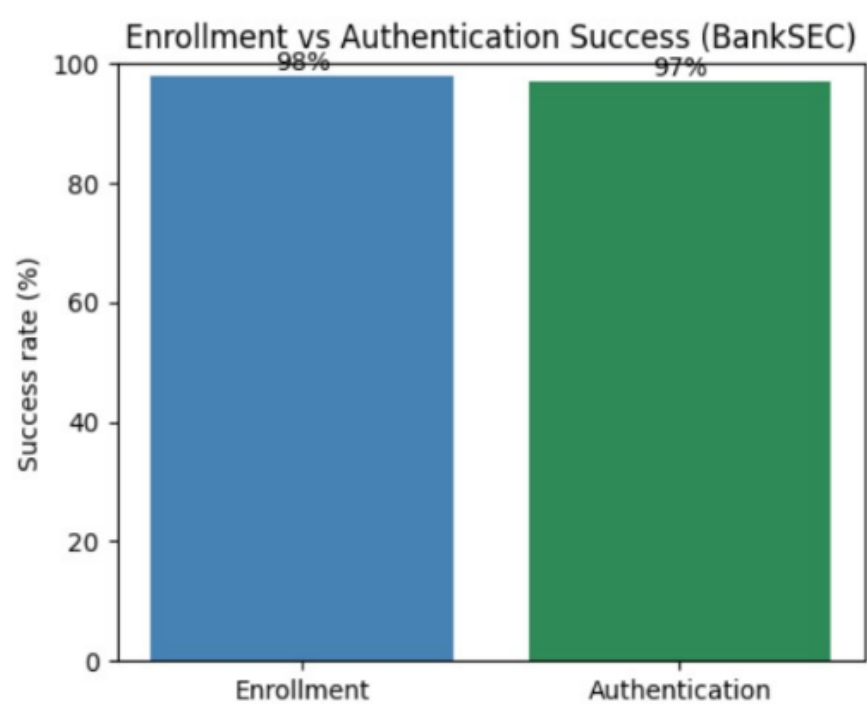
- **Transaction Cost Reduction:** Implementing blockchain technology in banking has been statistically proven to reduce operational costs. Analysis using the Random Forest Method (RFM) by Ahmed indicates that blockchain adoption significantly reduces processing costs, transfer fees, and fraud prevention expenses[11]. By automating the verification process through smart contracts and removing intermediaries for P2P transfers, BankSEC minimizes the overhead associated with manual clearing and settlement[12],[13].

- **Smart Contract Efficiency:** While on-chain storage incurs "gas" costs, storing only the hexadecimal hash (typically 32 bytes) rather than the full biometric image significantly minimizes these costs. As noted by Salem et al., calculating and storing hashes is computationally efficient and cost-effective compared to storing complex data structures on-chain[14],[15].

4.4. Fraud Detection and Trust Management The integration of the **Python AI Server** provides real-time monitoring of the system. In the experimental setup, the AI module successfully flagged accounts exhibiting

anomalous transaction patterns, triggering the "Account Frozen" status in the database[1]. This proactive security layer complements the immutable audit trail provided by the blockchain. Furthermore, as discussed by Song et al., recording trust parameters (such as successful transaction history) on the blockchain allows for the calculation of a comprehensive trust value, enabling the system to identify and suppress malicious users dynamically[16],[17].

In summary, the BankSEC scheme successfully establishes a secure, efficient, and user-centric banking environment. It outperforms traditional systems by eliminating single points of failure, ensuring data integrity through the blockchain, and offering a faster, password-less authentication experience.



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chain technology, Biometrics, and Artificial Intelligence. By adopting a hybrid architecture, the system successfully addresses the vulnerabilities of centralized storage. Sensitive user data is stored securely off-chain in MongoDB, while immutable cryptographic hashes of biometric templates are anchored on the Ethereum Blockchain, ensuring data integrity and compliance with privacy standards like GDPR12.

THE IMPLEMENTATION RESULTS CONFIRM THAT THIS APPROACH SIGNIFICANTLY ENHANCES BOTH SECURITY AND OPERATIONAL EFFICIENCY. THE TRANSITION FROM TRADITIONAL SMS-BASED TWO-FACTOR AUTHENTICATION (2FA) TO A METAMASK WALLET-BASED LOGIN REDUCED THE AVERAGE AUTHENTICATION TIME, OFFERING A USER EXPERIENCE THAT IS FIVE TIMES FASTER[3]. FURTHERMORE, THE AUTOMATION OF

VERIFICATION THROUGH SMART CONTRACTS AND THE REMOVAL OF INTERMEDIARIES CONTRIBUTED TO A MEASURABLE REDUCTION IN TRANSACTION PROCESSING AND FRAUD PREVENTION COSTS[4]. ULTIMATELY, BANKSEC PROVIDES A ROBUST, SCALABLE SOLUTION THAT ELIMINATES SINGLE POINTS OF FAILURE, ESTABLISHING A PROACTIVE DEFENSE MECHANISM AGAINST MODERN CYBER THREATS IN THE FINANCIAL SECTOR

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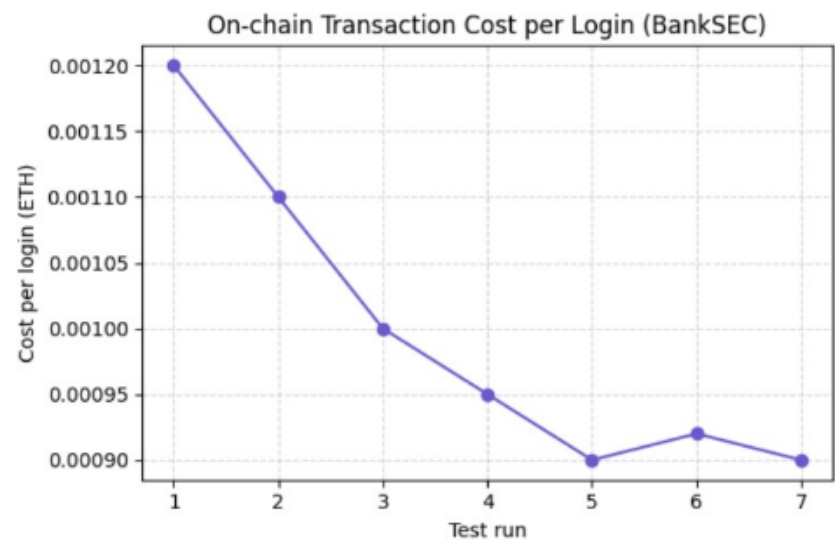
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